Project Plan; Design process of transmitting antenna

Group 5: [redacted] S. Letkiewicz

December 23, 2023

I. Introduction

March 23, 2023

A. Scope

The goal of the project is to define a set of essential parameters and their significance in the antenna performance. Given a set of conditions within which the antenna must operate, the justification of optimal characteristics and parameter values are provided. A structured project approach of analytical design, simulation, tweaking, and implementation cycle is described.

A signal is to be transmitted from the ground level to a receiving dipole antenna located approximately 30 meters away at an elevation of 50 meters above the ground.

The signal to be transmitted is generated by an AM transmitter module QAM-TX1 with an output impedance of 50Ω and a frequency of 433 MHz. The significance of some parameters such as directivity, signal gain, bandwidth, and polarization will be investigated.

II. Design

A. Specifications

1) Directionality and antenna gain: The signal is generated with QuasarUK AM transmitter module QAM-TX1. This is a transmission module that uses amplitude modulation (AM) in the form of amplitude shift keying (ASK). This is a method of transmission in which a signal is sent out in discretised amplitude levels to a reciever. These discrete amplitude differences represent different combination of binary code which can be read by the receiver and decoded for the specific application of the signal, such as, unlocking a garage door.

The data sheet of the transmitter module [1] presents its various specifications, one of which being a transmission range of 50 metres. With this information, the required directivity can be calculated.

Using the previously defined elevation and horizontal distance away from the dipole, the straight line distance to the dipole can be found. This was found to be 60 metres away from the transmitter. Looking at the data sheet, it can be seen that using only the transmitter module, the maximum distance for the transmitted signal to be measured (r_1) is 50 metres [1]. If the the same power is to be achieved at 60 metres (r_2) using an antenna, the gain can be expressed as a ratio between the two distances. This is then converted to decibels (dB) to get equation 1.

$$G_{Antenna} = 20 \cdot log_{10}(\frac{r_2}{r_1}) = 1.58dB \tag{1}$$

Furthermore, the antenna gain is also equal to the directivity (D_0) multiplied by the radiation efficiency (e_r) of the antenna. As the radiation efficiency of the antenna is extremely difficult to calculate, the antenna is assumed to be completely efficient and therefore at a value of 1 so as to get a rough estimate of the gain required. This means that the antenna gain is equal to the directivity of the antenna as seen in equation 2.

$$G_{Antenna} = D_0 \cdot\ e_r = D_0 \tag{2}$$

Using this equation, a value of approximately 1.58 dBi can be found for the required directivity in the transmission across a 60 metre distance. However, this value will likely lead to a weak signal despite it still being registrable. To ensure the working of the antenna with a strong enough signal, a value of at least 10 dBi will be the aim for the antenna directivity as this seems achievable [2]–[4]

2) Impedance: The transmitter module has a 50Ω output impedance, the input impedance of the antenna should thus match 50 Ω to avoid reflection and inefficiencies in the antenna.

3) Polarization: The receiving antenna is a dipole with linear polarization. The use of a circularly polarized antenna guarantees a loss of half of the power, which may be a necessary trade-off in the settings where parallel receiver-transmitter alignment is not possible. Given that the transmitter and receiver can both be accurately positioned with reference to the ground level, the use of a linearly polarized transmitter antenna is better suited.

4) Bandwidth: As the transmitter module uses amplitude modulation (AM) at 433 MHz and does not utilise frequency modulation, the bandwidth of the antenna doesn't have to be very large. However, it should still be large enough to account for errors in the AM-transmitter. Therefore, the bandwidth of the antenna should be a value of around 1 to 2% of the frequency.

5) Specifications summary: The specifications discussed have been summarised in the following table:

Specification	Value	Unit
Directivity	10	dBi
Antenna impedance	50	Ω
Bandwidth	2	%

B. *Alternative ideas and design choices*

There are many different antenna types whose properties are considered when crafted for particular applications. Some of the more known antenna types include the dipole, monopole, loop, array, and aperture [5]. The main disadvantages of the alternative antenna designs are summarised in table **??**.

Dipole antennas consist of two conductors which radiate in an omnidirectional fashion [6]. For this reason, the dipole antenna is not considered as a potential candidate for the project, as it lacks directionality. Monopole and loop antennas are also not considered for this application, as a more directional alternative will yield more efficient results.

On the other hand, array antennas are directive. Array antennas are a mixture of dipole, monopole and loop antennas that are arranged to work as one single antenna. For instance, Yagi–Uda antenna that consists of multiple half-wave dipoles, which radiate a beam along one driven element down the remaining ones **ANTENNATHEORYYAGIUDA**. The log-periodic antenna is also a type of array antenna that has a large directivity. The advantage of the log-periodic antenna over the Yagi-uda is its lager bandwidth [5], however the antenna is physically larger, making it less feasible in comparison.

Aperture antennas include the horn antenna and the parabolic dish antenna. An aperture antenna consists of a dipole or monopole surrounded by a larger structure used to guide the waves. As a result, these antennas have great directionality. The issue with an aperture antenna is its feasibility. For our specifications, the dish antenna would have to be three meters in length, which is not feasible for the preset budget of 25 euro. The horn antenna does not have this problem, however its dependence on the flare angle[1] [7] can prove to be a difficult manufacturing issue, therefore it has also been disregarded from the potential choices.

Lastly, the helical antenna, is a conducting wire wound in the form of a helix. This antenna does meet our desired specification and is achievable construction wise. The downside of this antenna is that it is circularly-polarized, causing it to lose approximately 50 % of the gain at the receiving linearly polarized dipole antenna. However, the antenna does not appear to have any major construction difficulties. From the antennas considered, the Yagi-uda and the helical have stood out as potential candidates for this application. Due to the downside of the helical being circularly-polarized, it will instead remain as a potential back alternative if construction of the Yagi-uda were to fail.

[1]Angle of the horn structure to the indented signal direction

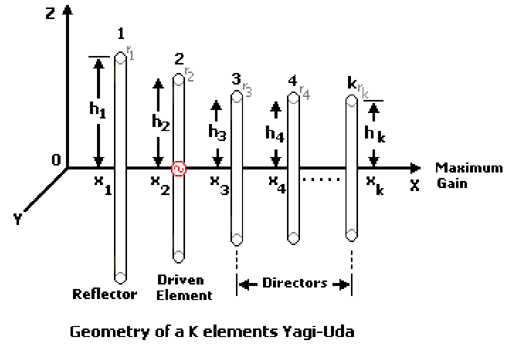


Fig. 1: Geometry of the Yagi-Uda

1) Yagi-Uda: The Yagi-Uda is a commonly used linearly polarised antenna. The antenna consists of the driven element, which often is a folded dipole, a reflector and depending on the required directivity, a range of directors. The reflector reflects the signal strength on the backside of the antenna which otherwise would have been present in the case of a regular dipole antenna. This reflector is typically 15% bigger than the driven component [2]. The directors in front of the antenna reflect and direct the signal in the desired direction of propagation usually in steps of 5%.

a) Pros:

- High directivity
- Linear polarization, since the receiving antenna shares this form of polarization, thus minimizing losses.
- Easy to construct.

b) Cons:

- Increasing the gain is difficult after a certain point (20 dB) since adding more directors yield diminishing returns and the antenna will become lengthy.

c) Frequency: Since the frequency that needs to be sent was determined to be 433 MHz, the antenna must be designed to work on this frequency. To calculate the desired wavelength the following formula will be used, where c is the speed of light

$$v = \lambda * f = c \tag{3}$$

After filling in this formula, the wavelength λ must be 0.6928m, rounded off to 4 decimals.

It is important that the length of the driven component must equal to $\frac{1}{2}(2n+1)$ times the wavelength, where n is a natural number. This is important since otherwise there will be gain losses due to the wave cancelling each other out if the length of the antenna is not half of the wavelength since the antenna can be approximated as a standing wave. Since the wavelength was determined to be 0.6928m, the approximate length of the driving element should be 0.3464m. The exact length of the antenna should be derived from further and simulations

2) Helical Antenna: The helical antenna in axial mode is a popular choice due to its circular polarization, relatively high gain and directivity [8] [9]. In contrast to a linearly polarized transmitter and receiver combination, where the transmitter and receiver have to be positioned parallel to each other for full power transmission, a circularized polarized receiving and transmitting antenna combination can transmit half the maximum available power independent of their spatial orientation.

A helical antenna can also be linearly polarized by specifically choosing its dimensions. This configuration is called normal mode, but this configuration also has a downside. In figure 2, where the radiation pattern of the normal mode is on the left and of the axial mode is on the right. It is evident from the radiation pattern of the normal mode that it has less directivity than the axial mode. The Yagi-Uda is also linearly polarized, however offers much higher directivity, which is required to transmit the data.

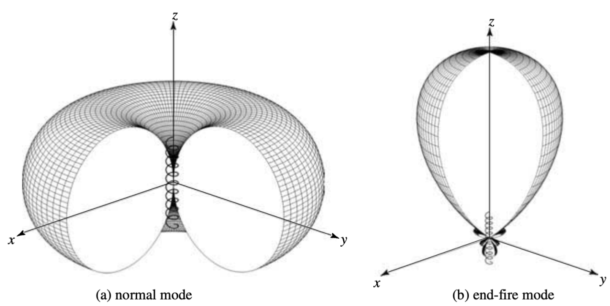


Fig. 2: Radiation pattern of the helical antenna in normal mode (left) and axial mode (right) [8]

C. Reflectors

Reflectors can improve the gain of an antenna in a specific direction[8]. Therefore, they are useful to implement in directional antennas, such as the antenna that is to be designed in this project. There are several types of reflectors with different characteristics, some of which can be seen in figure 3.

Due to the construction difficulty, only a couple more feasible designs are considered below.

1) Planar Reflectors: Planar reflectors are the easiest reflectors to construct and model. However, as shown in figure 3.a, the electromagnetic (EM) waves lose some of their directivity in the reflection.

2) Corner Reflectors: Corner reflectors are almost as simple to build as planar reflectors. The angle of the reflector as shown in figure 3.b has a significant impact. A sharp angle causes a lot of reflections which results in a lower gain as well as a little bit of noise due to the retardicity of the signal.

3) Curved Reflectors: Parabolic reflectors have a high gain and the EM fields are all parallel and highly directional, as shown in figure 3.c. Parabolic reflectors are difficult to

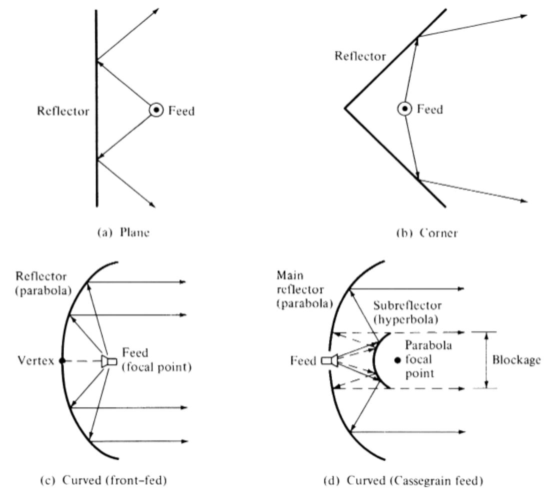


Fig. 3: Geometrical configuration for some reflector systems [8]

build and are generally considered not feasible within our project.

Antenna type	Why it is not chosen	Source
Loop	Impedance matching difficult, large size	[10]
Horn	Very large, Therefore out of budget	[11]
Microstrip Patch	Low directivity, could be large	[12]
Grid dish/ reflector	Too large, difficult to make	[13]
Dipole and monopole	Too low directivity	[14] [15]

TABLE I: Summary of alternative designs and their drawbacks.

D. Feasibility

In an earlier paragraph, the length of the driving element of the antenna was determined to be 0.3464m. This is not extraordinarily long or short and lies within the realms of possibility to build.

Antenna material has to meet the following criteria; high conductivity, low cost and structural integrity. Two options immediately come to mind for these specifications: copper and aluminium. They are both relatively inexpensive in the required amount, have great conductivity and are widely available.

To construct the antenna base, a non-conducting material is necessary. This is where the conducting elements will be attached to. A good option for this would be wood since wood is cheap and easy to work with.

Naturally, the antenna will not perform as intended by the design and simulation unless the signal power loss is minimized. The most prominent power cut may occur as a result of signal reflection due to the improper impedance matching. To maximize power transfer, the impedance of the transmitter, impedance of the coaxial cable and the antenna have to be equal. The coaxial cable are designed with a defined 50Ω impedance and so are most of the transmitters, an antenna will therefore have to be of a matching impedance. If the antenna design results in a smaller impedance, a matching network of inductors and capacitors can be used, with minimal power loss.

To conclude, the design of the Yagi-Uda antenna is feasible within the scope of this project.

III. Planning

A. Block Diagram

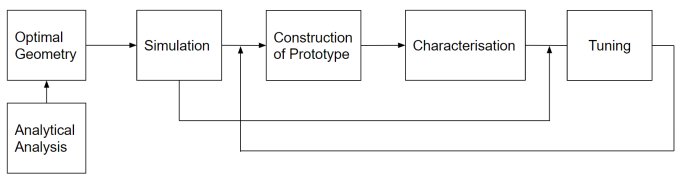


Fig. 4: Functional block diagram of the process of how the antenna will be designed and built with feedback

Analytical analysis and optimal geometry: Once the choice of antenna is narrowed to two options, an analytical approach will be taken to derive the required specifications of the antenna to meet the requirements of the transmission. From this, the optimal geometry including sizes and spacing will be derived to be later used in simulation. The other antenna will be kept as a potential back-up option in case some difficulties would arise.

Simulation and analysis of flexibility for a given design: Once the analytic behaviour of the antenna and the related optimal geometries are derived, the feasibility in tolerances for construction will be assessed alongside other set out specifications. Firstly the basic specifications of the antenna will be tested such as the directivity, impedance and the radiation pattern. Following this, small deviations from the ideal geometry will be introduced in the simulation, and by comparing the resulting directivity and other parameters, the required tolerances will be assessed to determine the feasibility of constructing the antenna. By comparing the data of the two antennas, a final choice can be made regarding whether a potential sacrifice in directivity or signal strength is worth having larger tolerances, hence easier construction. The material will also be taken into account and the impedance of the antenna will be approximated.

Construction of prototype: Once a design is chosen, and the geometry and tolerances are clearly defined the prototype will be constructed. The main purpose of the prototype is to check both the feasibility in construction as well as the ultimate performance of the antenna through characterisation.

Characterisation: To characterise the antenna it is important to identify and quantify the main elements that define performance such as gain, directivity, impedance and more.

To experimentally determine the gain, an experiment can be set up which uses another antenna with known characteristics as a base to compare the experimental antenna to. If the measured gain of the experimental antenna is bigger, this means that the power received from the transmitting experimental antenna will also be bigger. Formula 5 relates these two and will be discussed later.

The Friis transmission equation 4 can be used to calculate the path loss in free space where P_R is the power (Watt) received, P_T is the power (Watt) given to the transmitter, G_T is the antenna gain in the direction of the receiving antenna, G_R is the antenna gain in the direction of the receiving antenna, λ is the wavelength (m) and R (m) is the distance between the 2 antennas [16]

$$P_R = \frac{P_T G_T G_R \lambda^2}{4\pi * R^2} \tag{4}$$

To quantify the antenna gain performance, formula 5 can be used. In this formula, G_G is the gain of the known antenna and P_{R2} is the power received by the experimental antenna.

$$G_T = G_G * \frac{P_{R2}}{P_R} \tag{5}$$

If P_{R2} is bigger, the value of P_{R2} will increase, showing an increase in the gain. Subsequently, if P_{R2} is lower, the gain will be lower

To determine the impedance, a VNA can be used to determine the reactance and resistance of the antenna, in total which will yield the impedance, ideally of 50Ω.

Tuning: Once the antenna has been characterised and the gain and the impedance have been found, these parameters will be tuned to better match specifications. The simulation step will feed into this tuning step as the simulations will be used to determine the effects of changing certain geometries, and before any changes are made to the construction, the suggested new geometry will be simulated. This final step will involve a lot of back and forth between these two steps, and will with time result in a well-tuned antenna which therefore meets specifications.

B. Division of tasks and Team roles

The team has been divided into three main departments, design optimization which consists of Roman, Thorvald and Film, the construction team, consisting of Enan and Thimo, and lastly documentation with Evalds and Szczepan. The purpose of this split is to assign a specific tasks to each person without restricting anyone from working on different segments of the project. Overlapping in these roles will occur and is not discouraged, it is simply to account for all the necessary aspects in the project.

The design optimization team is in charge of optimizing the performance of the antenna, this includes choosing the material and geometric properties of the antenna, that will allow it to reach the desired specifications. The team will have to conduct simulations in COMSOL, as well as determining the overall antenna directivity, impedance and other parameters in MATLAB.

In the meantime, the construction team will concentrate on what type of cable is most suitable for the antenna, how to prevent as much power loss as possible. Once complete, the team will continue to sourcing the material necessary for the antenna and start the building process.

Lastly, the documentation team will move between the two departments, assisting where needed and overseeing the project. Another important role of this team is to document important design decision and report important finding in the report.

C. Time table and Process specifications

A timeline in the form of a Gantt chart can be seen in Fig. . The timeline takes into consideration exams and the holidays, giving the design team enough time to produce a well preforming design. The deadlines for the initial and final report have also been accounted for across the three teams depicted before.

IV. Process control

Monitoring the quality and controlling the progress for the group will be done by the documentation team. This is the main purpose behind having two designed people overlooking the project. Communication is also an important factor of the quality control. To tackle this problem, all group members will stay in contact with each other and also the TA.

[1] RF Solutions, *Rf solutions: Qam-tx1-433 - datasheet pdf tech specs*, version DSQAM-TX1-1, QuasarUK, Nov. 2005. [Online]. Available: https://www.datasheets.com/en/part-details/qam-tx1-433-rf-solutions-29653696#datasheet.

[2] P. Grajek, B. Schoenlinner, and G. Rebeiz, "A 24-GHz high-gain Yagi-Uda antenna array," *IEEE Transactions on Antennas and Propagation*, vol. 52, no. 5, pp. 1257–1261, May 2004, ISSN: 1558-2221. DOI: 10.1109/TAP.2004.827543.

[3] C. Chen and D. Cheng, "Optimum element lengths for Yagi-Uda arrays," *IEEE Transactions on Antennas and Propagation*, vol. 23, no. 1, pp. 8–15, Jan. 1975, ISSN: 1558-2221. DOI: 10.1109/TAP.1975.1141001.

[4] C. A. Balanis, *Modern Antenna Handbook*, en. John Wiley & Sons, Sep. 2011, Google-Books-ID: UYpV8L8GNCwC, ISBN: 9781118209752.

[5] P. J. Bevelacqua, *List of Antennas (Antenna Types)*, 2016. [Online]. Available: https://www.antenna-theory.com/antennas/main.php (visited on 03/23/2023).

[6] *Monopole and dipole antennas*, en-GB, Jan. 2021. [Online]. Available: https://enqt.de/en/news/monopole-and-dipole-antennas-characteristics-and-differences/ (visited on 03/23/2023).

[7] T. Agarwal, *Horn Antenna : Working, Types, Radiation Pattern & Its Applications*, en-US, Feb. 2022. [Online]. Available: https://www.elprocus.com/horn-antenna/ (visited on 03/23/2023).

[8] C. A. Balanis, *Antenna Theory Analysis and Design*. Wiley-Interscience, 2005, ISBN: 978-0471667827.

[9] tutorialspoint, *Antenna theory - helical*, en-US. [Online]. Available: https://www.tutorialspoint.com/antenna_theory/antenna_theory_helical.htm# (visited on 03/23/2023).

[10] P. Bevelacqua, *Loop antennas*, en-US. [Online]. Available: https://www.antenna-theory.com/antennas/smallLoop.php (visited on 03/23/2023).

[11] P. Bevelacqua, *The horn antenna*, en-US. [Online]. Available: https://www.antenna-theory.com/antennas/aperture/horn.php (visited on 03/23/2023).

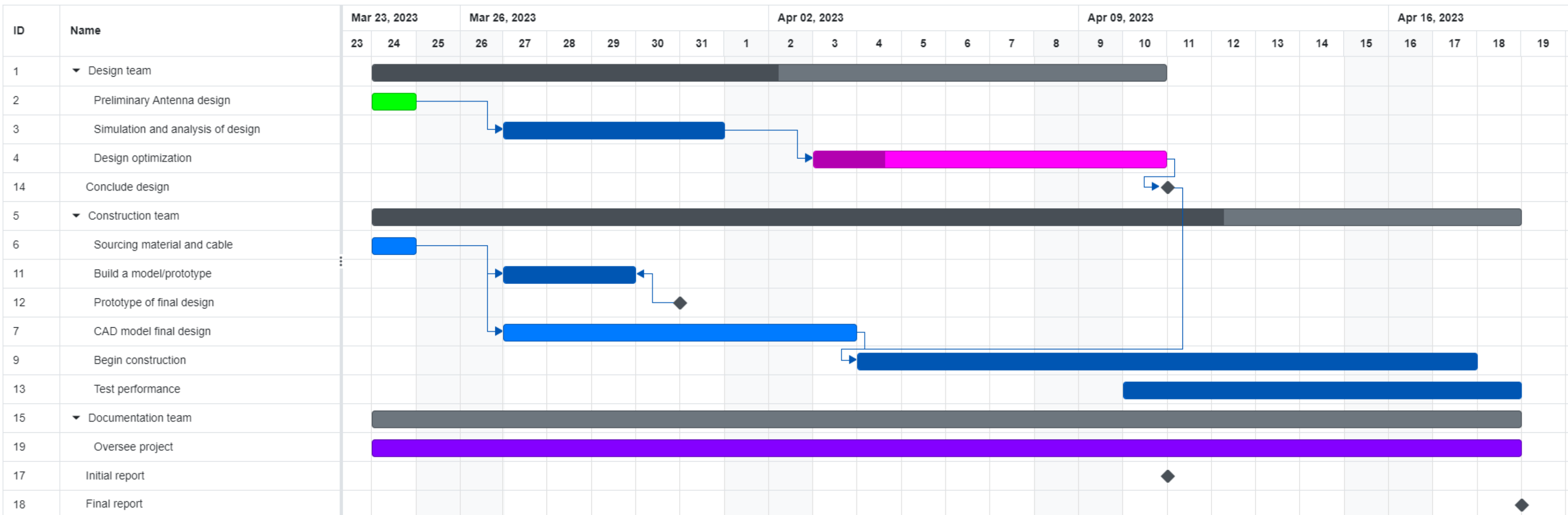
[12] P. Bevelacqua, *Microstrip (patch) antennas*, en-US. [Online]. Available: https://www.antenna-theory.com/antennas/patches/antenna.php (visited on 03/23/2023).

[13] P. Bevelacqua, *He parabolic reflector antenna (satellite dish)*, en-US. [Online]. Available: https://www.antenna-theory.com/antennas/reflectors/dish.php (visited on 03/23/2023).

[14] P. Bevelacqua, *The dipole antenna*, en-US. [Online]. Available: https://www.antenna-theory.com/antennas/dipole.php (visited on 03/23/2023).

[15] P. Bevelacqua, *The monopole antenna*, en-US. [Online]. Available: https://www.antenna-theory.com/antennas/monopole.php (visited on 03/23/2023).

[16] P. Bevelacqua, *Antenna-theory.com*.



ID
Name
Mar 23, 2023
Mar 26, 2023
Apr 02, 2023
Apr 09, 2023
Apr 16, 2023
23 24 25 26 27 28 29 30 31 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 2.
1 Design team
2 Preliminary Antenna design
3 Simulation and analysis of design
4 Design optimization
14 Conclude design
5 Construction team
6 Sourcing material and cable
11 Build a model/prototype
12 Prototype of final design
7 CAD model final design
9 Begin construction
13 Test performance
15 Documentation team
19 Oversee project
17 Initial report
18 Final report